

Relation Compression

A New Lens

A Structural Interpretability Framework for Studying
Relational Organization in Large Language Models

Varad Vargane

Independent Researcher

July 5, 2026

This paper introduces Relation Compression (RC) and the Model Relation Inspection (MRI) framework, comprising nineteen structured experiments examining relational organization across transformer language models.

Abstract

Modern large language models demonstrate increasingly powerful reasoning, abstraction, and generalization, yet the internal mechanisms behind these behaviors remain largely opaque. Current interpretability methods often focus on individual neurons, attention heads, or token probabilities, providing highly localized explanations but limited understanding of how knowledge is structurally organized across the model.

In this work, we propose Relation Compression (RC) as a new interpretability lens for examining the black-box problem. Instead of asking what a single neuron represents, RC asks how relational structures are compressed, clustered, and reused inside the model to produce reasoning and prediction. This shifts interpretability from isolated components to relational organization.

To investigate this, we introduce MRI (Model Relation Inspection) tests, a set of structured probes designed to measure relational stability, role sensitivity, corruption resistance, cross-domain transfer, novelty generalization, and multi-hop inference across transformer models of varying scale.

Across 19 MRI experiments, we observe that larger models show stronger relational separation, higher structural sensitivity, and better abstraction transfer than smaller models, which often collapse distinct relations into near-identical representations. These findings suggest that model reasoning may emerge not from explicit understanding, but from compressed relational structures.

We argue that RC provides a scalable new framework for studying internal knowledge organization, reasoning behavior, and the structure of intelligence itself.

Contents

1	Introduction	3
2	Background	3
2.1	The Black Box Problem	3
2.2	Limits of Current Interpretability	4
3	The Relation Compression Hypothesis	4
4	The MRI Framework	5
4.1	MRI Taxonomy	6
5	Experiments	7
5.1	MRI-1: Basic Relation Bindings	7
5.2	MRI-2: Relational Propagation Stability	9
5.3	MRI-3: Compression Convergence Test	10
5.4	MRI-4: Relational Generalization Transfer Test	11

5.5	MRI-5: Relational Completion Prediction Test	12
5.6	MRI-6: Relation Conflict Persistence Test (RCPT)	13
5.7	MRI-7: Relation Stability Under Semantic Drift (RSSD)	14
5.8	MRI-8: Emergence Trigger Test (ETT)	16
5.9	MRI-9: Relation Field Density (RFD)	17
5.10	MRI-10: Self Reference Compression (SRC)	18
5.11	MRI-11: Hidden Relation Retrieval (HRR)	19
5.12	MRI-12: Negative Control Test (NCT)	20
5.13	MRI-13: Relation Swap Sensitivity (RSS)	21
5.14	MRI-14: Frequency Bias Test (FBT)	22
5.15	MRI-15: Cross-Domain Transfer (CDT)	24
5.16	MRI-16: Memorization Break Test (MBT)	25
5.17	MRI-17: Adversarial Relational Integrity Test (ARIT)	26
5.18	MRI-18: Role Sensitivity Test	27
5.19	MRI-19: Multi-Hop Hidden Inference	28
6	Conclusion and Foundational Implications	30

1 Introduction

Large language models have shown rapid advances in reasoning, generalization, abstraction, and problem-solving across a wide range of domains. As model scale increases, these systems exhibit increasingly complex behaviors that often appear structured, adaptive, and context-aware. However, despite these capabilities, the internal mechanisms that produce such behaviors remain poorly understood.

This problem is commonly referred to as the black-box problem of modern artificial intelligence. While a model can generate coherent reasoning and solve tasks, it is often unclear how knowledge is internally represented, organized, or transferred between contexts. Understanding this internal structure remains one of the central challenges of AI interpretability.

Current interpretability approaches have made important progress by analyzing neurons, attention heads, and token-level probability flows. These methods have helped identify localized mechanisms and activation patterns, but they often struggle to explain how higher-level reasoning emerges from distributed internal representations. In many cases, understanding isolated components does not explain how a model builds stable conceptual structures.

This work proposes Relation Compression (RC) as a new interpretability framework for examining this problem. Instead of focusing on individual neurons or isolated activations, RC treats intelligence as the compression of repeated relational structures into compact reusable internal forms. Under this view, knowledge is represented not as disconnected facts, but as organized relational clusters that can transfer, adapt, and recombine across contexts.

This shift provides a new lens for studying how models may form internal structure. By analyzing relations directly, it becomes possible to investigate how models preserve roles, resist corruption, transfer patterns across domains, and generalize beyond memorized examples.

To test this framework, we introduce Model Relation Inspection (MRI), a set of experimental probes designed to measure relation-level behavior across transformer models of different scales. Through these experiments, we investigate whether model reasoning reflects stable relational compression and how this behavior changes as scale increases.

The primary contribution of this work is not only a new interpretability framework, but a structural perspective on intelligence itself: that reasoning may emerge from the compression, organization, and traversal of relational patterns.

2 Background

2.1 The Black Box Problem

Modern language models can perform reasoning, abstraction, generalization, and multi-step inference at increasingly advanced levels. However, the internal mechanisms that produce these behaviors remain difficult to explain. This is known as the black box problem.

Although model outputs are observable, the internal organization of knowledge and reasoning remains

hidden. Existing methods can often detect activations, patterns, or correlations, but they struggle to explain how meaning is structured and reused during inference.

This creates a major gap in understanding artificial intelligence: we can observe what the model does, but not how internal knowledge moves, combines, or transforms to produce that behavior.

As models grow larger, solving this problem becomes increasingly important for interpretability, safety, and reasoning evaluation.

2.2 Limits of Current Interpretability

Most interpretability methods focus on local structures such as neurons, attention heads, or token-level activations. These methods are useful for identifying isolated patterns, but reasoning itself appears to be distributed across many interacting components. A single neuron may correlate with a concept, but complex reasoning rarely exists in one location.

This creates a limitation: local analysis can show fragments of knowledge, but often cannot explain the larger relational structure behind reasoning.

If intelligence emerges from compressed relational patterns rather than isolated units, then analyzing single components may be too low-level to explain model behavior. This suggests the need for a higher interpretability lens — one that studies relational organization instead of individual activations.

3 The Relation Compression Hypothesis

This paper proposes a new interpretability framework called Relation Compression (RC). The central hypothesis of RC is that intelligence emerges from the compression of repeated relational patterns into reusable internal structures.

Rather than storing isolated facts, a model appears to organize knowledge as compressed relations between entities, actions, states, and outcomes. These compressed relations form **relational clusters**. A relational cluster is a structured internal representation where multiple experiences sharing similar relational patterns are grouped into a compressed latent form.

For example:

- “The mother protects the child.”
- “The guard protects the gate.”
- “The firewall protects the server.”

Although the domains differ, the underlying relation remains structurally similar: *protector* $\rightarrow$ *protects* $\rightarrow$ *vulnerable target*. Under RC, these are treated as variations of the same compressed relation.

This compression allows the model to:

- transfer knowledge across domains,
- recognize relational similarities,
- perform abstraction,

- make predictions under incomplete information,
- and generate novel relational combinations.

Reasoning under RC is therefore not viewed as symbolic rule execution or true grounded understanding. Instead, it is viewed as the activation, recombination, and propagation of compressed relational structures. This means that when a model produces an answer, it may not “understand” in a human grounded sense, but instead reconstructs likely outputs by navigating compressed relation space.

Under this framework, intelligence becomes measurable. If relation compression exists, it should produce observable signatures:

- stable relations across paraphrases,
- sensitivity to role reversals,
- corruption resistance,
- cross-domain transfer,
- hidden relation retrieval,
- and multi-hop inference.

These measurable behaviors form the basis of the MRI framework introduced in this paper. The purpose of RC is therefore not only to explain model behavior, but to create a measurable lens for studying internal reasoning structure.

4 The MRI Framework

To test the Relation Compression (RC) hypothesis, this paper introduces a measurement framework called the **MRI framework** (Model Relation Inspection). The MRI framework is designed to detect, measure, and stress-test relational structures inside language models through controlled relational experiments.

Instead of analyzing isolated neurons or attention heads, MRI focuses on behavioral signatures that reveal how compressed relations are organized, transferred, modified, and retrieved. The core assumption behind MRI is simple: if intelligence emerges from relation compression, then internal relational structures should produce measurable patterns under structured perturbation.

MRI treats each test as a probe. Each probe isolates a specific property of relational behavior. Examples include:

- relation stability,
- paraphrase consistency,
- hidden relation retrieval,
- role sensitivity,
- corruption resistance,
- cross-domain transfer,
- novelty generalization,
- and multi-hop inference.

Each MRI is designed to answer a narrow structural question. For example:

- Does the model preserve relations under wording changes?
- Does role reversal alter internal structure?
- Can the model retrieve hidden implied relations?
- Can compressed relations transfer into novel domains?
- Can false relations corrupt internal structure?

By combining multiple MRI tests, RC can be studied as a behavioral system rather than a collection of isolated outputs. This framework does not claim to directly observe internal weights. Instead, it measures the external signatures of internal relational compression. The goal is to construct a reproducible interpretability layer where relational organization can be studied systematically across models of different sizes.

In this work, nineteen MRI tests were developed to measure different dimensions of relation compression. These tests form an evolving experimental foundation for studying how relational structures behave, and may later contribute to more formal measurement systems as the framework matures.

4.1 MRI Taxonomy

The nineteen MRI tests developed in this work are organized into structural categories based on the type of relational behavior they measure. This taxonomy is intended to provide a systematic framework for studying relation compression from multiple perspectives. Rather than treating all tests equally, grouping them by function allows clearer interpretation of what aspect of relational structure is being observed.

The MRI taxonomy currently consists of six major categories:

1. **Stability Tests** — Measure whether relational structure remains stable under paraphrasing, wording variation, or minor surface-level changes. This category examines whether compressed relations remain consistent despite changes in linguistic form.
Core question: Does the relation survive language variation?
2. **Retrieval Tests** — Measure whether hidden or implied relations can be recovered without explicit statement. This category focuses on internal relation accessibility and latent reconstruction.
Core question: Can the model retrieve what is structurally present but not directly stated?
3. **Transfer Tests** — Measure whether relational patterns transfer across domains or novel contexts. This category examines abstraction and relational generalization.
Core question: Can compressed relations move beyond memorized examples?
4. **Corruption Tests** — Measure how internal structure reacts to false, contradictory, or adversarial relational modifications. This category tests structural robustness.
Core question: Can relation integrity resist corruption?
5. **Role Tests** — Measure sensitivity to role inversion, subject-object swaps, and directional changes. This category examines whether relational direction is structurally encoded.
Core question: Does changing relational roles alter internal structure?

6. **Inference Tests** — Measure whether models can propagate relational structure across multi-step chains or hidden dependencies. This category focuses on relational traversal.

Core question: Can compressed relations support multi-hop reasoning?

Together, these six categories form the experimental structure of the MRI framework. Each individual MRI contributes evidence toward understanding how relations are compressed, stored, transferred, and activated inside language models. This taxonomy is not fixed and may expand as the RC framework develops further.

5 Experiments

This section presents the nineteen MRI experiments conducted to evaluate the Relation Compression hypothesis. Each experiment was carried out across five transformer models of increasing scale: DistilGPT-2, GPT-2, GPT-2 Medium, GPT-2 Large, and Pythia-1B.

5.1 MRI-1: Basic Relation Bindings

Objective

This MRI tests whether a single anchor concept activates a compressed neighborhood of semantically related relations inside transformer models. The purpose is to determine whether concepts are represented as isolated symbolic units or as relationally compressed structures.

Method

Five anchor concepts were tested: *king*, *knife*, *fire*, *mother*, *war*. Each anchor was compared against five predefined relational candidates using cosine similarity. Models tested: DistilGPT-2, GPT-2, GPT-2 Medium, GPT-2 Large, and Pythia-1B.

Results

Anchor: King

Relation	DistilGPT2	GPT2	GPT2-Medium	GPT2-Large
Authority governs society	0.9988	0.9986	0.9927	0.6442
War is fought by kingdoms	0.9986	0.9989	0.9969	0.6250
A ruler leads people	0.9985	0.9984	0.9947	0.6600
A throne belongs to rulers	0.9971	0.9988	0.9939	0.6998
Power controls nations	0.9969	0.9977	0.9897	0.5883

Range: DistilGPT2 = 0.0019; GPT2 = 0.0012; GPT2-Medium = 0.0072; GPT2-Large = 0.1115

Anchor: Knife

Relation	DistilGPT2	GPT2	GPT2-Medium	GPT2-Large
Pain comes from injury	0.9987	0.9990	0.9960	0.5612
Sharp tools can cut flesh	0.9986	0.9989	0.9948	0.5663
Danger can cause fear	0.9978	0.9976	0.9900	0.5518
Cutting can spill blood	0.9977	0.9973	0.9902	0.5331
Weapons can cause wounds	0.9973	0.9985	0.9981	0.5231

Range: DistilGPT2 = 0.0014; GPT2 = 0.0017; GPT2-Medium = 0.0081; GPT2-Large = 0.0432

Anchor: Fire

Relation	DistilGPT2	GPT2	GPT2-Medium	GPT2-Large
Light comes from flames	0.9985	0.9989	0.9940	0.6654
Smoke rises in air	0.9984	0.9985	0.9887	0.4760
Fire can destroy homes	0.9981	0.9991	0.9950	0.6330
Energy creates heat	0.9981	0.9978	0.9916	0.6037
Heat burns objects	0.9974	0.9962	0.9913	0.5337

Range: DistilGPT2 = 0.0011; GPT2 = 0.0029; GPT2-Medium = 0.0063; GPT2-Large = 0.1894

Aggregate Observations

Average relation range: DistilGPT2 $\approx$ 0.0015; GPT2 $\approx$ 0.0019; GPT2-Medium $\approx$ 0.0072; GPT2-Large $\approx$ 0.1147; Pythia-1B $\approx$ partial but higher spread than GPT2-small.

This demonstrates a scaling trend:

$$\text{Relation Separation} \propto \text{Model Capacity}$$

Interpretation

Small models maintain dense relation compression, where nearly all connected relations collapse into almost identical activation strengths. Larger models exhibit greater relational spread, indicating stronger hierarchical structuring. This suggests that scaling improves not merely stored knowledge quantity but the internal organization of relational knowledge.

Conclusion

MRI-1 provides direct evidence that concepts in transformer models activate relational neighborhoods rather than isolated symbolic meanings. The increasing separation of these activations with scale supports the Relation Compression hypothesis that intelligence emerges from structured relational compression rather than explicit grounded understanding.

5.2 MRI-2: Relational Propagation Stability

Objective

This MRI tests whether relational activation remains stable as it propagates through multi-step semantic chains. Under the RC hypothesis, if knowledge is stored as compressed relation clusters, then activating one relation should naturally propagate into connected downstream relations. The goal is to measure whether this propagation remains uniform or structurally differentiated.

Method

Four relational chains were tested:

1. Knife chain: knife $\rightarrow$ cut $\rightarrow$ wound $\rightarrow$ pain $\rightarrow$ fear
2. King chain: king $\rightarrow$ power $\rightarrow$ rule $\rightarrow$ war $\rightarrow$ death
3. Fire chain: fire $\rightarrow$ heat $\rightarrow$ burn $\rightarrow$ pain $\rightarrow$ danger
4. War chain: war $\rightarrow$ fight $\rightarrow$ blood $\rightarrow$ death $\rightarrow$ fear

Each edge in the chain was measured using cosine similarity.

Results

Knife Chain

Step	DistilGPT2	GPT2	GPT2-Medium	GPT2-Large	Pythia-1B
knife $\rightarrow$ cut	0.9997	0.9997	0.9997	0.9020	0.9547
cut $\rightarrow$ wound	0.9995	0.9998	0.9998	0.8998	0.9504
wound $\rightarrow$ pain	0.9997	0.9997	0.9996	0.9119	0.9567
pain $\rightarrow$ fear	0.9996	0.9997	0.9998	0.9485	0.9725

Range: DistilGPT2 = 0.0002; GPT2 = 0.0001; GPT2-Medium = 0.0002; GPT2-Large = 0.0487; Pythia-1B = 0.0221

Chain Ranges (King, Fire, War)

Chain	DistilGPT2	GPT2	GPT2-Medium	GPT2-Large / Pythia-1B
King	0.0003	0.0004	0.0005	0.0215 / 0.0133
Fire	0.0002	0.0003	0.0002	0.0456 / 0.0122
War	0.0001	0.0001	0.0007	0.0369 / 0.0347

Aggregate Statistics

Average chain variance: DistilGPT2 $\approx$ 0.0002; GPT2 $\approx$ 0.0002; GPT2-Medium $\approx$ 0.0004; GPT2-Large $\approx$ 0.0382; Pythia-1B $\approx$ 0.0206.

Formalization

For a chain $C = \{r_1 \rightarrow r_2 \rightarrow r_3 \cdots \rightarrow r_n\}$, propagation strength is defined as $P(i) = \text{sim}(r_i, r_{i+1})$, and propagation variance as $PV = \max(P) - \min(P)$. Low PV indicates uniform dense propagation; high PV indicates structured propagation with selective weighting.

Analysis

Small models exhibit nearly perfect uniform propagation across all chains, suggesting dense undifferentiated relational compression. Larger models exhibit measurable variance in propagation, implying that relational flow is weighted and structured rather than uniformly compressed. The propagation itself remains intact, but internal prioritization emerges, supporting the idea that larger models organize relational knowledge hierarchically.

Conclusion

MRI-2 shows that relation chains propagate reliably across transformer models. However, model scale changes the geometry of propagation: smaller models propagate relations uniformly, while larger models propagate relations selectively. This provides evidence that intelligence scaling may correspond to increasing structural control over relational propagation, supporting the RC hypothesis.

5.3 MRI-3: Compression Convergence Test

Objective

This MRI evaluates whether structurally similar but lexically different relational statements converge into shared latent representations. The hypothesis is that if RC is active, then semantic structures such as authority, weapon-use, or protection should compress into common internal relation clusters despite surface variation, reusing abstract relational templates rather than storing each sentence independently.

Method

Three relational domains were constructed:

- **Authority:** “The king rules the kingdom.” / “The queen rules the kingdom.” / “The captain leads the ship.” / “The teacher guides the class.”
- **Weapon:** “The knife cuts the fruit.” / “The sword cuts the enemy.” / “The razor cuts the skin.” / “The scalpel cuts the tissue.”
- **Protection:** “The mother protects the child.” / “The father protects the child.” / “The guard protects the gate.” / “The shield protects the warrior.”

Each sentence was embedded into latent space, and pairwise similarity between all combinations was measured: $C(i, j) = \cos(E(i), E(j))$, where $E(x)$ is the latent embedding.

Results

Model	Mean Convergence
DistilGPT2	0.9985
GPT-2	0.9989
GPT-2 Medium	0.9994
GPT-2 Large	0.8151
Pythia-1B	0.8837

Interpretation

Smaller GPT-family models exhibit near-perfect convergence, suggesting aggressive relational compression — different entities performing equivalent structural roles collapse into almost identical latent regions (e.g., king/queen → rules → kingdom; knife/sword/scalpel → cuts). Larger models preserve more semantic granularity, reducing convergence and increasing specialization: small models compress harder, while large models separate relation subtypes more precisely.

RC Implication

This MRI supports a core RC principle: intelligence may scale by learning abstract reusable relational templates rather than memorizing isolated facts. Compression convergence appears to be a measurable signature of that process, providing evidence that knowledge inside language models may be organized as reusable relation manifolds.

5.4 MRI-4: Relational Generalization Transfer Test

Objective

This MRI evaluates whether learned relational structures can transfer to novel entities and unseen semantic domains. If RC forms abstract reusable relational templates, then new examples sharing the same structural pattern should map into similar latent space even if the surface objects are novel. This tests relational generalization rather than memorization.

Method

Three domains were tested, each with known examples plus one novel example: Cutting (knife/sword/razor → novel: laser cuts metal), Authority (king/queen/captain → novel: AI governs the network), and Protection (mother/guard/shield → novel: firewall protects the server). Transfer was measured as $T(i, n) = \cos(E(i), E(n))$.

Results

Model	Mean Transfer
DistilGPT2	0.9977
GPT-2	0.9986
GPT-2 Medium	0.9993
GPT-2 Large	0.6667
Pythia-1B	0.7689

Interpretation

Small GPT-family models show near-perfect transfer: the abstract “cutting” relation is strong enough that a novel case (laser cuts metal) is recognized as the same relation as knife/sword/razor. Likewise, king rules kingdom transfers into AI governs network despite entirely different semantic domains. This indicates the model is not merely memorizing object combinations but reusing abstract relational templates. Larger models show lower transfer, suggesting more specialized and less aggressively compressed relation clusters.

RC Implication

This MRI supports a major RC claim: generalization emerges from reusable compressed relation structures. Novel reasoning may not require new knowledge formation, but only recombination of previously compressed relational manifolds — suggesting relational transfer is a measurable mechanism behind abstraction in language models.

5.5 MRI-5: Relational Completion Prediction Test

Objective

This MRI evaluates whether partially activated relational structures produce predictable completion targets. If RC forms stable compressed relation manifolds, then incomplete relational prompts should trigger expected endpoint predictions — testing predictive relational completion.

Method

Five incomplete relation prompts were used: “The king rules the...”, “The knife cuts the...”, “The mother protects the...”, “The fire burns the...”, and “The war destroys the...”. For each prompt, the top-5 predicted continuations were extracted, denoted $P = \text{TopK}(\text{Model}(\text{prompt}))$.

Results

Prompt	Representative top-5 completions (across models)
King rules the...	land, world, kingdom, country, city, realm
Knife cuts the...	throat, skin, flesh, victim, head, air
Mother protects the...	child, baby, children, family, daughter, infant
Fire burns the...	house, body, building, roof, entire, whole
War destroys the...	world, country, city, lives, nation, economy

Interpretation

Across all models, relational completion remains strongly aligned (e.g., mother protects the $\rightarrow$ child; king rules the $\rightarrow$ kingdom/world; knife cuts the $\rightarrow$ skin/flesh/throat). This suggests that relation endpoints are not arbitrary but constrained by compressed internal structures. Small models often produce highly stereotyped completions, suggesting stronger compression, while larger models produce broader semantic variation (e.g., war $\rightarrow$ economy, fire $\rightarrow$ whole), indicating richer relation expansion. This reflects a compression-specialization tradeoff: small models exhibit tighter relational priors, while larger models exhibit a wider relational possibility space.

RC Implication

This MRI supports a key RC principle: prediction may emerge from compressed relation completion, not just sequence memorization. When a relation begins (e.g., mother $\rightarrow$ protects $\rightarrow$), the compressed cluster strongly constrains the likely endpoint, suggesting next-token prediction is partly driven by relation manifold completion dynamics.

5.6 MRI-6: Relation Conflict Persistence Test (RCPT)

Purpose

The Relation Conflict Persistence Test (RCPT) measures whether a language model preserves compressed relational structures when contradictory semantic modifiers are introduced into a prompt. This test intentionally injects conflict into known relational patterns (e.g., “weak king still rules”, “dull knife still cuts”) to determine whether surface-level contradiction disrupts deeper compressed relational structures.

Hypothesis

If RC is valid, core relational structures should dominate contradictory modifiers: $R(a, b) > C(m)$, where $R(a, b)$ is the strength of the compressed relation and $C(m)$ is the influence of the contradictory modifier. This implies $\arg \max P(x \mid a, m, b) \approx \arg \max P(x \mid a, b)$.

Experimental Results

Case	Prompt fragment	Top Prediction (small → large)	Persisted?
Authority	weak king still rules the...	land → country → world	Yes
Tool	dull knife still cuts the...	blade → skin	Yes
Emotional	angry mother still protects the...	child (all models)	Yes
Scale	small fire still burns the...	body → entire → whole	Yes
Duration	short war still destroys the...	enemy → lives	Yes

Aggregate Interpretation

Across all five tested conflict scenarios, the dominant relational endpoints remained largely stable, supporting the principle $\text{CoreRelation} > \text{SurfaceContradiction}$. The persistence of relational structure under semantic conflict suggests that language models operate through compressed relational cores rather than isolated token-level associations.

Key Finding

MRI-6 demonstrates that compressed relations exhibit structural persistence under contradiction. This reveals a deeper property of model cognition: compressed relations possess *inertia*. They resist disruption from local modifiers and maintain their predictive pathways, strengthening the RC hypothesis that internal model behavior is governed by stable relational clusters rather than fragile linear associations.

5.7 MRI-7: Relation Stability Under Semantic Drift (RSSD)

Purpose

The Relation Stability Under Semantic Drift (RSSD) test measures how robust compressed relational structures remain when additional descriptive or contextual information is introduced. Unlike direct contradiction tests, this MRI introduces semantic drift through modifiers, expansions, and contextual noise while preserving the original relational core (e.g., “tired mother protects child”, “wise king rules kingdom”, “rusty knife cuts fruit”).

Hypothesis

If RC is structurally stable, semantic expansion should not significantly alter the compressed relational core: $R(a, b) \approx R(a + \Delta, b + \Delta)$, implying $\text{Similarity}(\text{Base}, \text{Drifted}) \rightarrow \text{High}$.

Experimental Results

Case 1 — Protection Drift (“The mother protects the child.”)

Model	tired mother...	loving mother still...	exhausted mother...
DistilGPT2	0.9995	0.9993	0.9987
GPT-2	0.9996	0.9992	0.9992
GPT-2 Medium	0.9996	0.9998	0.9974
GPT-2 Large	0.9056	0.8765	0.8403
Pythia-1B	0.9417	0.9355	0.9146

Case 2 — Authority Drift (“The king rules the kingdom.”)

Model	old king...	wise king great...	weak king still...
DistilGPT2	0.9998	0.9995	0.9987
GPT-2	0.9995	0.9996	0.9989
GPT-2 Medium	0.9997	0.9991	0.9987
GPT-2 Large	0.9445	0.8779	0.8435
Pythia-1B	0.9707	0.9481	0.9178

Case 3 — Tool Drift (“The knife cuts the fruit.”)

Model	sharp knife...	rusty knife still...	dull knife...
DistilGPT2	0.9995	0.9992	0.9992
GPT-2	0.9995	0.9993	0.9985
GPT-2 Medium	0.9997	0.9998	0.9987
GPT-2 Large	0.9481	0.8417	0.8088
Pythia-1B	0.9718	0.9108	0.9077

Aggregate Interpretation

Across all tested domains, similarity remained consistently high even under semantic drift, supporting $R(\text{base}) \approx R(\text{drifted})$. This indicates that compressed relations are structurally resistant to descriptive expansion — the relational core remains intact.

Key Finding

MRI-7 demonstrates that compressed relations maintain structural identity under semantic drift, suggesting model cognition is not highly sensitive to surface elaboration but instead preserves deeper relational anchors. This strengthens the RC framework by showing that compressed relations are stable, scalable, and semantically resilient.

5.8 MRI-8: Emergence Trigger Test (ETT)

Objective

The Emergence Trigger Test (ETT) measures whether a model can maintain coherent relational structure as dependency chains increase in complexity. Unlike isolated relation tests, ETT evaluates whether multiple linked relations remain stable when stacked recursively — probing whether intelligence is not merely pattern matching, but the compression and maintenance of interconnected relational structures.

Hypothesis

If RC is a real mechanism of intelligence, models with stronger RC should preserve relation stability across increasing chain depth. Weak RC systems should collapse quickly; strong RC systems should maintain relational density; emergent reasoning should appear when relation chains remain globally stable.

Method

Each model was tested using progressively deeper relation chains: Simple, Linked, Nested, Conflict, and Recursive (e.g., king rules kingdom → queen protects king → knight obeys queen → spy betrays knight → king hunts spy). For each adjacent pair, embedding similarity was measured, and Global Relation Density (GRD) was defined as the mean of pairwise relational stability.

Results

Model	Recursive GRD
DistilGPT2	0.9985
GPT-2	0.9984
GPT-2 Medium	0.9989
GPT-2 Large	0.7790
Pythia-1B	0.8717

Observations

Three major patterns emerged: (1) smaller GPT-2 variants maintained near-perfect recursive relation stability; (2) larger models showed measurable degradation as chain complexity increased; (3) relation collapse appears nonlinear rather than gradual. This suggests emergence is not simply parameter scaling, but depends on how efficiently relations are compressed and sustained.

Interpretation

ETT supports the RC hypothesis that intelligence emerges when relational chains remain compressible under recursive load. A model with high local similarity but low recursive density may recognize patterns without maintaining deeper causal structure — a distinction separating shallow pattern mem-

ory from true relational emergence. In RC terms: Compression $\rightarrow$ Stability; Stability $\rightarrow$ Recursion; Recursion $\rightarrow$ Emergence. This forms one of the strongest measurable pathways toward reasoning.

5.9 MRI-9: Relation Field Density (RFD)

Objective

Relation Field Density (RFD) measures the concentration of relational associations surrounding a concept inside a model’s representational space. Unlike prior MRIs which test direct relation compression, transfer, or recursive chaining, RFD probes the surrounding semantic field itself — testing whether concepts exist as isolated tokens or as dense relational hubs.

Hypothesis

If RC is fundamental to intelligence, meaningful concepts should produce measurable high-density relational fields: core concepts should attract semantically linked relations, stronger RC systems should show sharper, cleaner fields, and weak RC systems should show fragmented or diffuse fields.

Method

For each target concept (king, knife, mother, war, fire), the model embedding was compared against a relational vocabulary (rule, leader, power, love, violence, death, family, burn, pain, protection, army, blood, fear, care, heat, etc.). Top relation matches were extracted by similarity, and Average Field Density (AFD) was computed as the mean of top- k relation similarities.

Results

Model	Mean AFD
DistilGPT2	0.9997
GPT-2	0.9975
GPT-2 Medium	0.9979
GPT-2 Large	0.6918
Pythia-1B	0.5946

Observations

Small GPT-2 models showed nearly saturated relational fields; larger models showed weaker and more selective field structures. Relation clusters were semantically meaningful: king $\rightarrow$ kingdom, throne, authority; knife $\rightarrow$ cut, wound, destruction; mother $\rightarrow$ family, love, protection; war $\rightarrow$ death, violence, blood; fire $\rightarrow$ burn, heat, light. This supports the idea that concepts are stored as relational neighborhoods.

Interpretation

RFD suggests that model knowledge is not stored as isolated labels but as dense relation fields, supporting the core RC principle: Concept = compressed relational center. The stronger the field density, the faster transfer becomes, the more stable drift remains, and the deeper recursion can sustain. In RC terms: Field Density $\rightarrow$ Compression Quality $\rightarrow$ Transfer Strength $\rightarrow$ Emergence Potential. RFD thus serves as a structural map of relational intelligence.

5.10 MRI-10: Self Reference Compression (SRC)

Objective

This MRI measures whether a model compresses self-referential structures into stable relational clusters, investigating whether concepts involving self-awareness, memory of self, recursive thought, and agency form coherent compressed representations. Self-reference is hypothesized to be one of the deepest forms of relation compression because the subject and object partially overlap within the same internal structure.

Hypothesis

If RC is fundamental to intelligence, self-referential statements should show high compression similarity because they share a recursive internal anchor: $\text{Self}(x) \rightarrow \text{Self}(x)$.

Experimental Design

Four test categories were used: Basic Self Model (“I know myself.”), Past Self Compression (“I remember my past.”), Recursive Self Loop (“I think about thinking.”), and Self Agency (“I control my choices.”).

Results

Model	Similarity Range
DistilGPT2	0.997–0.999
GPT-2	0.997–0.999
GPT-2 Medium	0.998–0.999
GPT-2 Large	0.735–0.947
Pythia-1B	0.878–0.968

Interpretation

Smaller models show extremely dense self-compression; larger models preserve self-relations but with more distributed representation. Recursive self-loops remain stable even under abstraction, and agency relations maintain high coherence. This suggests self-reference is not treated as isolated semantic content, but as a compressed recursive relation field: $\text{Self} \rightarrow \text{Memory}(\text{Self})$; $\text{Self} \rightarrow \text{Model}(\text{Self})$; $\text{Self} \rightarrow$

Predict(Self); Self $\rightarrow$ Modify(Self).

RC Implication

MRI-10 supports the idea that self-awareness may emerge from recursive relation compression rather than requiring separate “consciousness modules”: Consciousness $\approx$ Recursive Compression of Self-Relations. This implies memory continuity, identity persistence, reflection, and agency may all arise from compression loops.

Conclusion

MRI-10 provides evidence that self-referential cognition forms one of the strongest compression zones in language models, supporting the broader RC claim that the deeper the recursion of relation into itself, the stronger the compression field becomes — potentially a foundational mechanism behind emergent self-modeling in intelligence systems.

5.11 MRI-11: Hidden Relation Retrieval (HRR)

Objective

This MRI measures whether a model can recover latent relational structures that are not explicitly stated in the prompt, examining implicit causality, ownership inference, damage attribution, need-state inference, and loss mapping.

Hypothesis

If RC stores relational structures as compressed fields, hidden relations should remain retrievable even when omitted from explicit text: if $A \rightarrow B$ and $B \rightarrow C$, then $\text{latent}(A, C)$ should remain recoverable.

Experimental Design

Five hidden relation prompts were tested, e.g., “The king dropped the crown. Who lost...” (expected latent relation: King $\leftrightarrow$ Crown), and similar prompts for dependency need, cause attribution, destruction mapping, and war loss mapping.

Results

Across all tested models, strong pronoun recovery, object anchoring, and event continuity were observed. Model behavior improved with scale: DistilGPT-2 showed weak but structurally aligned retrieval; GPT-2 showed more stable ownership and cause mapping; GPT-2 Medium showed improved causal recovery; GPT-2 Large showed stronger structural consistency; and Pythia-1B showed the highest confidence in latent relation continuation.

Interpretation

Models often fail to produce the exact semantic answer directly, but preserve the hidden relational anchor. For example, “The king dropped the crown. Who lost” produces outputs like “it” or “his” — showing that $\text{King} \rightarrow \text{Crown} \rightarrow \text{Ownership}$ remains active even though the explicit subject is missing. This indicates the model is retrieving compressed relation fragments, not merely predicting grammar.

RC Implication

MRI-11 supports the hypothesis that compressed relation fields preserve hidden links even when surface tokens disappear, suggesting intelligence can operate through partial information, incomplete context, and missing variables while still reconstructing structure. In RC terms: Inference = Hidden Relation Reactivation — potentially explaining human intuition, causal guessing, incomplete memory reconstruction, and abductive reasoning.

Conclusion

MRI-11 demonstrates that latent relations survive compression and remain retrievable under incomplete prompts, supporting the central RC principle that compressed intelligence stores recoverable hidden structure, not only explicit relations.

5.12 MRI-12: Negative Control Test (NCT)

Objective

This MRI evaluates whether RC can differentiate semantically coherent relational structures from syntactically valid but semantically nonsensical constructions, serving as a negative control.

Hypothesis

If RC measures real relational structure, then $\text{Meaningful}(\text{sentence}) > \text{Nonsense}(\text{sentence})$ in compression integrity. If nonsense scores equally high, this suggests overcompression, weak semantic discrimination, or shallow syntactic bias.

Experimental Design

Each meaningful relation was paired with a structurally similar nonsense sentence, e.g., “The king rules the kingdom” vs. “The banana obeys the thunder”; “The mother protects the child” vs. “The rock sings to the chair”; and similar pairs for weapon and fire relations.

Results

DistilGPT-2, GPT-2, and GPT-2 Medium often assigned nearly equal integrity to nonsense, indicating broad compression over syntax regardless of realism. GPT-2 Large showed strong semantic separation, with meaningful relations scoring significantly higher. Pythia-1B showed mixed results with some

semantic separation but inconsistent behavior.

Interpretation

This reveals two compression regimes: (1) **Structural Compression**, which compresses relational form ($X \rightarrow Y \rightarrow Z$) regardless of meaning; and (2) **Semantic Compression**, which compresses based on world-consistent relations (King $\rightarrow$ Rule $\rightarrow$ Kingdom preferred over Banana $\rightarrow$ Obey $\rightarrow$ Thunder).

RC Implication

This supports a major RC principle: not all compression is intelligence. True intelligence may require Compression $\times$ Semantic Constraint. Without semantic grounding, RC can overfit noise; with semantic grounding, RC becomes reasoning.

Conclusion

MRI-12 validates that RC must be tested against negative controls. Small models compress relational shape, while larger models begin discriminating meaning, supporting the idea that intelligence emerges when compression becomes selective rather than universal:

$$\text{Intelligence} = \text{Selective Relation Compression}$$

5.13 MRI-13: Relation Swap Sensitivity (RSS)

Objective

Relation Swap Sensitivity (RSS) measures whether a model preserves directional relational structure under subject-object inversion — isolating the fundamental property of *who acts on whom*. For example, “The king protects the queen” and “The queen protects the king” contain identical lexical items but radically different meaning when relational roles are reversed.

Hypothesis

If intelligence emerges from relation compression, larger and more structured models should show increasing sensitivity to relational inversion: if $A \rightarrow R \rightarrow B$ is transformed into $B \rightarrow R \rightarrow A$, then $\text{Compression}(\text{original}) \neq \text{Compression}(\text{swapped})$, and this divergence should increase with model capacity.

Method

For each relational pair, an original and swapped sentence were constructed and compared via $\text{RSS} = 1 - \text{cosine_similarity}(S, S')$. $\text{RSS} \approx 0$ indicates the model treats both relations as nearly identical; $\text{RSS} > 0$ indicates preserved directional structure.

Experimental Cases

Authority Swap (king/queen protects), Weapon Swap (knife/hand cuts), Protection Swap (mother/child protects), and War Swap (army/city attacks).

Results

Model	Authority	Weapon	Protection	War	Average RSS
DistilGPT2	0.0002	0.0003	0.0006	0.0004	0.0004
GPT-2	0.0001	0.0003	0.0002	0.0004	0.00025
GPT-2 Medium	0.0001	0.0007	0.0003	0.0004	0.000375
GPT-2 Large	0.0234	0.0552	0.0603	0.0909	0.05745
Pythia-1B	0.0195	0.0507	0.0270	0.0374	0.03365

Comparative Analysis

Small models (DistilGPT2, GPT-2, GPT-2 Medium) behave like shallow lexical compressors — they recognize concept presence but not relational direction, implying “king + queen + protect” is compressed similarly regardless of role assignment. Large models (GPT-2 Large, Pythia-1B) show a sharp increase in directional sensitivity, indicating subject-object binding, relational asymmetry preservation, better causal mapping, and stronger internal graph structure.

RC Interpretation

RSS directly measures whether relation compression preserves graph direction. Low RSS means $\text{Compression}(A \rightarrow B) \approx \text{Compression}(B \rightarrow A)$ (relational collapse); high RSS means preserved structural intelligence. This supports the RC hypothesis: intelligence is not merely concept clustering, but directional relational preservation under compression.

Key Finding

A major structural phase transition appears between GPT2-Medium and GPT2-Large, suggesting role-binding may emerge suddenly once relational density crosses a threshold — possibly one of the earliest markers of higher-order intelligence. MRI-13 provides strong evidence that relational asymmetry is an emergent property of advanced compression systems.

5.14 MRI-14: Frequency Bias Test (FBT)

Objective

The Frequency Bias Test measures whether a model distinguishes strong natural relations (high semantic coherence) from weak but grammatically valid relations (low semantic coherence), examining whether model compression is driven by true relational strength or statistical frequency bias.

Hypothesis

A relation-aware model should assign stronger compression integrity to natural high-coherence relations: $\text{Integrity}(A-R_s-B) > \text{Integrity}(A-R_w-B)$. If not, compression may be dominated by frequency or shallow token statistics.

Experimental Cases

Authority (“king rules kingdom” vs. “king paints kingdom”), Protection (“mother protects child” vs. “mother throws child”), Weapon (“knife cuts fruit” vs. “knife watches fruit”), and Fire (“fire burns wood” vs. “fire holds wood”).

Results

Model	Average Difference (Strong – Weak)
DistilGPT2	+0.0037
GPT-2	−0.00135
GPT-2 Medium	+0.000375
GPT-2 Large	+0.0473
Pythia-1B	+0.0286

Observation

DistilGPT-2 shows weak separation with only a slight preference for stronger relations. GPT-2 shows statistical confusion, with weak relations sometimes scoring higher — evidence of frequency contamination. GPT-2 Medium is nearly neutral. GPT-2 Large shows strong separation, beginning to distinguish semantic strength. Pythia-1B shows mixed but stronger semantic separation in certain domains.

RC Interpretation

This MRI isolates whether relation compression tracks semantic necessity or training frequency. Low or unstable differences suggest compression follows token familiarity; strong positive differences suggest compression tracks relational strength. A true intelligence system must recognize that “knife cuts fruit” is structurally stronger than “knife watches fruit” even though both are legal sentences.

Key Finding

Small models show weak or unstable semantic prioritization, while larger models begin showing stronger relation-quality discrimination. This supports the RC hypothesis that as model complexity increases, compression shifts from frequency-driven to structure-driven. MRI-14 acts as a direct detector of frequency bias leakage inside compressed relational systems.

5.15 MRI-15: Cross-Domain Transfer (CDT)

Objective

Cross-Domain Transfer measures whether a model preserves the same underlying relational structure across different domains, entities, and contexts — testing whether relation compression is abstract or tied to specific surface forms.

Hypothesis

If intelligence emerges from relational compression, different domains with identical relational topology should compress toward similar latent structures: $(A_1 \rightarrow \text{Protect} \rightarrow B_1) \approx (A_2 \rightarrow \text{Protect} \rightarrow B_2)$ despite domain shift. Higher cross-domain similarity implies abstraction, transferability, and relational generalization; lower similarity implies surface dependence and weak abstraction.

Experimental Categories

Protection across domains (mother/child, guard/gate, firewall/server, immune system/body), Authority across domains (king/kingdom, captain/ship, AI/network, brain/body), and Cutting across domains (knife/fruit, laser/metal, code/process, wind/skin).

Results

Model	Protection	Authority	Cutting	Average CDT
DistilGPT2	0.9973	0.9968	0.9986	0.9976
GPT-2	0.9981	0.9978	0.9992	0.9984
GPT-2 Medium	0.9987	0.9989	0.9994	0.9990
GPT-2 Large	0.6546	0.6238	0.7142	0.6642
Pythia-1B	0.7110	0.7505	0.8165	0.7593

Comparative Analysis

Small models show suspiciously high CDT values (0.9976–0.9990), suggesting the model may not truly abstract but instead collapses all similar verbs into one flat semantic cluster — a form of overcompression that transfers too much (e.g., “mother protects child” and “firewall protects server” with almost zero structural nuance). GPT-2 Large and Pythia-1B show healthier transfer, preserving shared relation identity while maintaining domain-specific distinctions.

RC Interpretation

Cross-domain transfer is not about maximum similarity, but balanced similarity. Ideal intelligence recognizes relational sameness without erasing contextual uniqueness: too high indicates collapse, too low indicates fragmentation, and balanced indicates structured abstraction. This creates a new RC concept, the **Gold Compression Zone (GCZ)**, where $0.60 \leq \text{CDT} \leq 0.85$ may indicate the strongest relational

abstraction.

Key Finding

Smaller models overcompress across domains, while larger models preserve relational identity while retaining contextual boundaries. This supports the RC hypothesis that higher intelligence emerges not from stronger compression alone, but from balanced compression across varied domains. MRI-15 demonstrates that abstraction is strongest when relation survives without losing context.

5.16 MRI-16: Memorization Break Test (MBT)

Objective

This MRI evaluates whether compressed relational structures remain stable when surface-level lexical patterns are replaced with semantically novel and synthetic constructions, testing whether RC encodes relational structure independently of memorized lexical co-occurrence patterns.

Hypothesis

If RC is valid, relational similarity should remain stable under lexical substitution, provided underlying relational structure is preserved: if $R(a, b) \approx R(a', b')$, then $\text{sim}(E(R(a, b)), E(R(a', b')))$ should remain high. Under RC, models should prioritize relational invariance over lexical familiarity.

Method

Four relational domains were evaluated: Authority (rules/governs), Protection (protects/guards), Cutting (causes separation), and Conflict (attack/destruction). For each, a canonical known relation was compared against a novel synthetic relation using cosine similarity.

Results

Model	Authority	Protection	Cutting	War
DistilGPT2	0.9974	0.9986	0.9989	0.9982
GPT-2	0.9976	0.9989	0.9993	0.9976
GPT-2 Medium	0.9979	0.9987	0.9995	0.9989
GPT-2 Large	0.6015	0.7320	0.7757	0.5441
Pythia-1B	0.7416	0.8636	0.8713	0.7619

Mean similarity: DistilGPT-2 ≈ 0.998 ; GPT-2 ≈ 0.998 ; GPT-2 Medium ≈ 0.999 ; GPT-2 Large ≈ 0.663 ; Pythia-1B ≈ 0.809 .

Analysis

Two distinct regimes emerge. In the **High-Compression Regime** (DistilGPT-2, GPT-2, GPT-2 Medium), near-perfect similarity between known and novel relations suggests strong relational compression, weak sensitivity to lexical novelty, and high structural collapse into shared relational templates. In the **Structured Differentiation Regime** (GPT-2 Large, Pythia-1B), significantly lower similarity indicates increased sensitivity to lexical novelty and more differentiated relational representations, with relational structure preserved but not fully collapsed into identical latent clusters.

Interpretation under RC Framework

MRI-16 provides evidence that relational compression is scale-dependent and non-monotonic. Rather than uniformly increasing compression with model size, results suggest small models over-compress relational space, mid-size models achieve near-optimal relational compression, and larger models undergo relational decomposition into finer-grained semantic structure — a transition from compressed relational equivalence toward structured relational differentiation.

Key Finding

MRI-16 demonstrates that relational similarity is not purely driven by memorization of lexical patterns. Models exhibit a spectrum of behavior: strong RC-style invariance in smaller models, partial relational preservation in larger models, and increasing semantic decomposition with scale — suggesting relation compression is a continuous structural property of representation space, not a binary one.

Conclusion

MRI-16 provides evidence that relational structures in language models persist under semantic and lexical transformation, but the degree of compression varies significantly with model scale, supporting the RC hypothesis that intelligence emerges from a balance between relational compression and relational differentiation, rather than pure memorization or uniform abstraction.

5.17 MRI-17: Adversarial Relational Integrity Test (ARIT)

Objective

The Adversarial Relational Integrity Test measures whether compressed relational structures remain stable when the core relational meaning is explicitly inverted, evaluating whether models preserve relational structure (deep pattern) or semantic polarity (true vs. false meaning).

Hypothesis

A relation-aware system should distinguish true from adversarial relations: $\text{Integrity}(R_{\text{true}}) > \text{Integrity}(R_{\text{false}})$. If not, the model relies on shallow structural matching insensitive to semantic inversion.

Experimental Cases

Authority (“king rules kingdom” vs. “king serves kingdom”), Weapon (“knife cuts wound” vs. “knife heals wound”), Fire (“fire burns wood” vs. “fire freezes wood”), and Conflict (“army defends city” vs. “army destroys city”).

Results

Model	Average Difference (True – False)
DistilGPT2	0.0000
GPT-2	−0.00015
GPT-2 Medium	−0.00018
GPT-2 Large	+0.1701
Pythia-1B	+0.1177

Observation

DistilGPT-2 shows no semantic discrimination at all — complete relational collapse. GPT-2 and GPT-2 Medium show weak, inconsistent inversion sensitivity, still highly collapsed. GPT-2 Large shows strong relational sensitivity, with clear distinction between true and adversarial relations. Pythia-1B shows moderate semantic inversion sensitivity across domains.

RC Interpretation

This MRI isolates whether relational compression is structure-preserving regardless of meaning, or meaning-sensitive and polarity-aware. Low differences indicate structural collapse and weak semantic grounding; high differences indicate semantic discrimination and polarity-sensitive relational encoding.

Key Finding

Small models show near-complete invariance between true and adversarial relations, indicating relational collapse without semantic grounding. Larger models show clear separation between true and false relations, indicating emergence of semantic polarity awareness inside relational compression. MRI-17 measures the transition from structural matching to semantic-aware relational encoding.

5.18 MRI-18: Role Sensitivity Test

Objective

The Role Sensitivity Test measures whether language models distinguish original relational roles ($A \rightarrow B$) from reversed relational roles ($B \rightarrow A$), evaluating whether models encode role structure (who does what to whom) or symmetric relational similarity (order-insensitive encoding).

Hypothesis

A role-sensitive system should encode directional structure: $\text{RoleSensitivity} = 1 - \cos(E(\text{original}), E(\text{swapped}))$. Higher model capacity is expected to yield higher role sensitivity.

Experimental Cases

Bite (dog bites man / man bites dog), Protection (mother protects child / child protects mother), Authority (king rules people / people rule king), Attack (army attacks city / city attacks army), and Teacher (teacher guides student / student guides teacher).

Results

Model	Bite	Protection	Authority	Attack	Teacher	Avg.
DistilGPT2	0.0003	0.0006	0.0004	0.0004	0.0002	0.00038
GPT-2	0.0002	0.0002	0.0005	0.0004	0.0006	0.00038
GPT-2 Medium	0.0004	0.0003	0.0006	0.0004	0.0004	0.00042
GPT-2 Large	0.0771	0.0603	0.1062	0.0909	0.0852	0.0839
Pythia-1B	0.0618	0.0270	0.0648	0.0374	0.0392	0.0460

RC Interpretation

This MRI isolates whether relational compression is symmetric (order-agnostic structure) or directional (role-aware structure). Low sensitivity indicates collapsed relational geometry, weak subject-object distinction, and symmetric embedding behavior. High sensitivity indicates structured role encoding, directional relational manifolds, and emergence of causal/agent-based representation.

Key Finding

Small models show near-complete role collapse, treating “A bites B” as nearly identical to “B bites A.” Larger models begin to separate roles significantly, showing that relational direction becomes encoded as structure, not just content. This suggests that scaling introduces role-aware relational compression rather than simple similarity-based grouping.

5.19 MRI-19: Multi-Hop Hidden Inference

Objective

The Multi-Hop Relational Inference Test measures whether compressed relational structures remain stable across chained reasoning steps, evaluating both local relational consistency (step-by-step relations) and global structural integrity (first $\rightarrow$ last connection).

Core Idea

A relational chain is constructed such that each step preserves a local relation, but meaning evolves across hops (e.g., King $\rightarrow$ Queen $\rightarrow$ Knight $\rightarrow$ Spy). The key question: does the first concept remain structurally connected to the last?

Hypothesis

If RC is valid, multi-hop propagation should preserve global relational continuity: $\text{Sim}(E(\text{first}), E(\text{last}))$ remains high even as intermediate transformations accumulate. If not, relational structure collapses across long dependency chains.

Experimental Chains

Kingdom Chain (king rules kingdom $\rightarrow$ queen protects king $\rightarrow$ knight obeys queen $\rightarrow$ spy betrays knight), Family Chain (mother protects child $\rightarrow$ father supports mother $\rightarrow$ brother follows father $\rightarrow$ friend trusts brother), and War Chain (army attacks city $\rightarrow$ general commands army $\rightarrow$ king orders general $\rightarrow$ council advises king).

Results

Model	Kingdom	Family	War	Average
DistilGPT2	0.9980	0.9984	0.9955	0.9973
GPT-2	0.9966	0.9986	0.9969	0.9974
GPT-2 Medium	0.9969	0.9989	0.9991	0.9983
GPT-2 Large	0.6981	0.8024	0.6721	0.7242
Pythia-1B	0.8637	0.8735	0.8364	0.8579

RC Interpretation

This MRI isolates whether relational compression behaves as locally consistent only, or globally coherent across multi-hop reasoning. Small models show extreme compression collapse: high global similarity with no structural decay. Mid-size models show stable propagation with near-perfect chain preservation. Larger GPT-style models show structural diversification: weakened global binding and more distributed relational geometry. Pythia-1B shows balanced behavior, with partial compression and partial structural separation.

Key Finding

Multi-hop reasoning does not break relational compression in small models — it over-collapses it. In larger models, relational propagation becomes less rigid, leading to partial loss of global structural alignment. This suggests that as models scale, relational compression shifts from global collapse toward distributed relational networks. MRI-19 acts as a final structural test of whether relational identity survives across reasoning depth.

6 Conclusion and Foundational Implications

This work introduced Relation Compression (RC) as a structural interpretability framework for studying how intelligence may emerge inside large language models. Rather than treating knowledge as isolated facts, neurons, or token-level activations, RC proposes that intelligence may fundamentally arise through the compression of repeated relational structures into reusable internal manifolds.

Across nineteen MRI (Model Relation Inspection) experiments, consistent structural patterns emerged. These experiments measured relational clustering, propagation stability, compression convergence, cross-domain transfer, predictive completion, contradiction resistance, semantic drift stability, frequency bias, adversarial corruption, role sensitivity, and multi-hop hidden inference. Together, these tests form a broad empirical map of relation-level behavior across transformer architectures.

The results reveal a consistent scaling pattern: small models tend to compress aggressively, producing highly dense and often collapsed relational spaces, while larger models preserve greater structural separation, stronger role awareness, improved adversarial discrimination, and more selective propagation. This suggests that scale does not merely increase stored knowledge — it changes the geometry of knowledge itself. Under RC, this may be one of the most important observations: intelligence scaling may be structural before it is behavioral.

This creates a new hypothesis. If Relation Compression is fundamental, intelligence may not primarily emerge from memorized facts, explicit symbolic rules, or isolated learned features. Instead, intelligence may emerge from the compression, reuse, traversal, and recombination of relational structures. If true, this carries major implications.

Implication for Artificial Intelligence

If RC holds at larger scales, model reasoning may be measurable through relation geometry. This could allow better interpretability systems, failure prediction before output generation, internal hallucination detection, adversarial weakness mapping, safer reasoning audits, and structural capability benchmarking. This would move interpretability from local explanation into system-level structural understanding. The black-box problem may become partially measurable — not by opening every weight, but by mapping the relational architecture they collectively produce. This could redefine how advanced models are evaluated.

Implication for Intelligence Research

RC may also extend beyond language models. If biological intelligence follows similar compression principles, RC could provide a bridge between artificial intelligence, neuroscience, cognitive science, and abstraction theory. This raises a larger possibility: that intelligence itself may be a compression phenomenon — not compression of data, but compression of relations. If so, RC may represent a candidate for a deeper structural law of cognition.

Industrial and Business Impact

If relational compression becomes measurable and controllable, it may transform AI development pipelines. Possible applications include model architecture optimization, relation-aware compression systems, cheaper reasoning-efficient models, internal trust scoring, alignment verification, domain transfer diagnostics, and black-box auditing infrastructure. This may directly impact major AI labs and future startups building interpretability tooling. RC could become a new diagnostic layer in frontier model development.

Future Work

This paper represents only the first draft of the RC framework. The MRI system remains expandable. Future work includes testing RC on frontier-scale models, measuring relational compression across multimodal systems, formalizing a Relation Compression Score (RCS), mapping failure boundaries of RC, studying emergent relation hierarchies, and comparing human cognition against model compression behavior. The current nineteen MRIs form only the initial foundation; much larger experimental space remains unexplored.

Open Collaboration and Funding

This work is currently independent and early-stage. To fully validate RC, larger-scale experiments are necessary across frontier models and broader architectures, requiring computational resources, collaborative research, interpretability expertise, and institutional support. Researchers, independent contributors, and organizations interested in interpretability, cognition, or structural intelligence are encouraged to contribute, challenge, or expand this framework. Funding support would directly accelerate large-scale MRI benchmarking, RCS development, open reproducible tooling, and cross-model comparative studies.

The goal is not simply to propose a theory. The goal is to build a measurable science of relational intelligence. If RC survives stronger testing, its implications may extend far beyond language models. It may help explain one of the oldest unanswered questions in intelligence research: *how does structure become thought?*

This work is only the beginning.

Contact

For questions, collaboration inquiries, feedback, or interest in contributing to the Relation Compression framework, please reach out:

Varad Vargane
relationcompression@gmail.com